1 Basics

This famous Einstein's mass-energy equation, $E = m.c^2$ should be rendered inline. However the following equation of motion should be rendered in a block:

$$v = u + a.t$$

2 Fractions & Roots

$$x_i = \frac{\sqrt{a^2 + b_i^2}}{c_i}$$

3 Symbols & Math-fonts

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \quad \forall x \in \mathbf{R}\ \exists\ \varepsilon > 0 \Rightarrow |x| < \varepsilon$$

$$\mathcal{F}(\theta) = \sum_{k=0}^{n} \binom{n}{k} \theta^k (1-\theta)^{n-k}$$

4 Calculus

$$\frac{du}{dx} + \frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

$$\int_0^1 (3x^2 + 2x + 1)\, dx$$

5 Multi-line equations

5.1 Centering & alignment

$$x = 5 \tag{1}$$
$$x + y = 3 \tag{2}$$

$$x = 5 \tag{3}$$
$$x + y = 3 \tag{4}$$

5.2 Multiline equation

$$(x+y+z)^3 = (x+y+z).(x+y+z).(x+y+z) = x^3+y^3+z^3+ \\ 3x^2y+3xy^2+3y^2z+3yz^2+3x^2z+3xz^2+6xyz \quad (5)$$

6 Matrices

$$\begin{matrix} 1 & 2 \\ 3 & 4 \end{matrix} \qquad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \qquad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$